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Dt:- 18/10/2025

25FEBS101IAEI25

Seat No. : 35

Student Name : Prachi

St. John College of Engineering and Management, Palghar, Maharashtra
Autonomous Institute
Internal Assessment Examinations, October 2025
AS & H - SEMESTER I
25FEBS101 - Engineering Mathematics – I

ALDEL EDUCATION TRUST

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)State the condition that defines an orthogonal matrix.	1	CO1	L2
	b)Evaluate $\frac{\partial^2}{\partial y \partial x}[\sqrt{xy}]$	1	CO2	L1
	c)How is the Jacobian matrix of a function $F(x,y) = (u(x,y), v(x,y))$ defined?	1	CO3	L1
	d)State the condition for a **unique solution** of a non-homogeneous system.	1	CO1	L2
	e)Evaluate $\frac{\partial^2}{\partial y^2}(1-2xy)^2$	1	CO2	L1
	f)State the condition for a point to be a saddle point.	1	CO2	L1
Q2	a)Find the rank of $A = \begin{pmatrix} 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 8 \end{pmatrix}$ by reducing to echelon form.	5	CO1	L3
	OR			
	b)Find the values of a,b and c if $A = 1/9 \begin{pmatrix} -8 & 4 & a \\ 1 & 4 & b \\ 4 & 7 & c \end{pmatrix}$ is orthogonal.	5	CO1	L3
Q3	a)If $u = log(tanx + tany)$ prove that $sin2x\frac{\partial u}{\partial x} + sin2y\frac{\partial u}{\partial y} = 2$.	5	CO2	L3
	OR			
	b)If $z = (3xy - y^3) - (y^2 - 2x)^{\frac{3}{2}}$, show that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.	5	CO2	L3
Q4	a)If $u = log(\frac{x^3+y^3}{x^2+y^2})$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1$.	5	CO3	L3
	OR			
	b)If $u = x^2 tan^{-1}(\frac{y}{x}) + y^2 sin^{-1}(\frac{x}{y})$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial y \partial x} + y^2\frac{\partial^2 u}{\partial y^2} = 2u$.	5	CO3	L3